Gliomas

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Author: ScholarRx

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Describe the typical clinical features of astrocytomas and the histologic features used for diagnosis.
2. Explain how astrocytomas are graded, and list the features that characterize glioblastoma (grade IV).
3. Describe the most common age and location for pilocytic astrocytoma, describe the characteristic histologic features, and explain the prognosis.
4. Describe the unique histologic and molecular features of oligodendroglioma that distinguish it from astrocytoma.
5. Compare and contrast the presentation of ependymoma in children and adults, describe the histologic features, and describe the typical clinical course in both children and adults.

CASE CONNECTION

TC is a 76-year-old male brought in after a seizure at home. His wife, BC, reports, "He'd been having headaches, but he didn't make anything of them. He just took something for it and went on with his work. He's been putting in long hours." TC has no significant medical history. His vital signs are normal. He is still difficult to arouse, consistent with a post-ictal state. He responds to noxious stimuli. Pupils are equal and reactive, and reflexes are intact though diminished. The remaining exam is normal. TC becomes more awake over about 20 minutes. You order an initial workup, including ECG, bedside glucose, blood tests, and a head CT scan. BC says, "The seizure was frightening. I thought he was dying."

What are the most likely causes of TC's seizure and headaches? Consider your answer as you read, and we'll revisit TC at the end of the brick.

GO TO CONCLUSION

What Are Gliomas?

Gliomas are tumors that arise from glial cells in the brain—astrocytes, oligodendrocytes, and ependyma and microglial cells. Accordingly, the three types of gliomas are astrocytomas, oligodendrogliomas, and ependymomas. About 3 of 10 of all brain tumors are gliomas, and most fast-growing brain tumors are gliomas. Their treatment often requires a combination of neurosurgical interventions, radiation therapy, and chemotherapy.

What are the four types of glial cells from which brain tumors can arise?

The World Health Organizations (WHO) classification system for gliomas is the most commonly used and consists of four grades in ascending order of benign to malignant. The highest-grade field seen on any mass will dictate grading for the whole neoplasm:

- Grade I: Pilocytic astrocytoma

INSTRUCTOR NOTE

> This is an example of a WHO grade I glioma
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- Grade II: Infiltrating low-grade astrocytoma

INSTRUCTOR NOTE

> This can also be referred to as a diffuse glioma and also is just one example of a WHO grade II glioma.
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- Grade III: Anaplastic astrocytoma
- Grade IV: Glioblastoma

What Is a Pilocytic Astrocytoma?

An astrocytoma is a tumor that arises from astrocytes, the star-shaped cells that serve as the supporting tissue for the brain. Pilocytic astrocytoma is a low-grade form of astrocytoma.

Incidence and Location

Pilocytic astrocytoma (WHO grade I) is the most common benign central nervous system (CNS) neoplasm in children, accounting for nearly 20% of all pediatric brain tumors. Peak occurrence is typically at age 8-13 years. It can also occur in adults (mean age of 20-25 years), and there is some association with neurofibromatosis type 1 in less common locations such as the optic chiasm. Pilocytic astrocytomas are typically found in the supratentorial region of the cerebrum in adults and have a predilection for cerebellar hemispheres in children. These tumors grow very slowly, usually progressing for about 2-4 years before they are diagnosed.

STRUCTOR NOTE

Classically, pilocytic astrocytomas are described as cystic with a mural nodule, especially in the cerebellum

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Histology

Pilocytic astrocytomas are considered biphasic tumors, with some sections of tightly packed bipolar cells with Rosenthal fibers and others with more loosely packed multipolar cells and microcysts. Rosenthal fibers are intracytoplasmic, brightly eosinophilic extensions with a corkscrew-like appearance (Figure 1). These are characteristic and high yield for this pathology. Pilocytic astrocytomas stain positive for glial fibrillary acidic protein (GFAP), because all astrocytes contain GFAP.

STRUCTOR NOTE

Please note that the presence of Rosenthal fibers are not just limited to pilocytic astrocytomas.

Also please note that all gliomas are positive for GFAP including Glioblastomas, which are described below

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QUIZ

Tap image for quiz

Figure 1

What is the typical histological feature of a pilocytic astrocytoma?

Presentation and Prognosis

Like the symptoms of most mass-occupying lesions, the symptoms of pilocytic astrocytomas are produced by the tumor location. Given its predilection for the cerebellum, common symptoms of an astrocytoma may include gait difficulty, ataxia, dizziness, and nystagmus. A child with new-onset clumsiness is a common presentation.

The prognosis in adults is slightly less favorable than in children, but the 5-, 10-, and 20-year survival rates are 85%, 80%, and 75%, respectively. In children, the 5-year progression-free survival rate after surgical resection is 90%-95%.

What Are Infiltrating Low-Grade Astrocytomas?

TRUCTOR NOTE

Again, sometimes the term Diffuse astrocytoma is used.

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Infiltrating low-grade gliomas are WHO grade II tumors. These gliomas are often astrocytic in nature but may also arise from oligodendrocytes.

Incidence and Location

These tumors make up about 10%-15% of gliomas in the adult population and have an incidence of 1.4 cases per 1 million people per year. The median diagnosis age is 30-40 years with a male-to-female ratio of 1:5. In adults, the majority of these astrocytomas occur in the supratentorial region, preferentially in the frontal and temporal lobes.

What is the typical presentation (age, sex, location) for a patient with infiltrating low-grade astrocytoma?

Histology

The definition for low-grade gliomas is hypercellular glial lesions with nuclear atypias but low mitotic activity, that is, a low degree of cell division. All grades of astrocytomas stain positive for GFAP because GFAP is a marker of astrocytes themselves. Infiltrating astrocytomas also stain positive for vimentin, an intermediate filament.

Presentation and Prognosis

Infiltrating low-grade astrocytomas often present with new-onset seizures in a previously healthy young adult.

The median survival for a patient diagnosed with infiltrating low-grade astrocytoma is highly variable, 2-12 years. One of the most important factors determining prognosis is patient age. Patients are usually treated with combination chemoradiation. Most experts recommend maximum safe debulking. Patients are never considered truly cured. Recurrence and progression are common, typically within 2 cm of the original tumor location.

What Are Anaplastic Astrocytomas?

Anaplastic astrocytomas are WHO grade III astrocytomas. Anaplastic describes poorly differentiated cells, and these tumors have a tendency to become malignant.

Incidence and Location

The incidence of anaplastic astrocytomas is only about 0.48 per 100,000 person/years. The mean age at diagnosis is 45 years with a slight male predominance. These tumors can arise anywhere in the CNS, including the brainstem and spinal cord, with a preference for the frontal and temporal lobes.

Histology

Histologically, these appear similar to the grade II infiltrating low-grade astrocytomas but with greater cellular atypia and more frequent mitotic figures (Figure 2). These tumors lack microvascular proliferation and necrosis.

TRUCTOR NOTE

These last two features are characteristic of WHO grade IV glioblastoma, see below

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QUIZ

Tap image for quiz

Figure 2

What is the histology of an anaplastic astrocytoma?

Presentation

Presentation largely depends on the exact location of the tumor, but most patients debut with seizures. However, many symptoms also stem from increased intracranial pressure produced as the tumor grows. Typical signs include headaches, focal deficits, and altered mental status. Tumors in the frontal lobe can cause memory problems, personality changes, and paralysis of the contralateral extremities. Tumors in the temporal lobes can cause seizure and speech and memory problems.

Prognosis

The treatment option of choice is surgical debulking of as much of the tumor as safely possible. If surgery is not possible, radiation alone is a treatment option. For adult patients, chemotherapy can also be used, but there is no approved chemotherapy regimen for children. The 5-year survival rate is about 23.6%. Elderly patients are more likely to die in the first year after diagnosis compared with young adults. The typical medial survival is 2-3 years.

What Are Glioblastomas?

Glioblastomas (also known as glioblastoma multiforme [GBM]) are WHO grade IV astrocytomas. Of all glial tumors, these are the most invasive; they grow rapidly and often invade surrounding tissues. These tumors are often devastating and rapidly fatal.

Incidence and Location

The incidence is 2-3 per 100,000 cases per year. These tumors account for more than half (52%) of all malignant primary brain tumors and 17% of both primary and metastatic brain tumors. Glioblastomas usually arise from hemispheric white matter typically in the frontal and temporal lobes. Sometimes they cross the midline via the corpus callosum or anterior commissure and invade the contralateral hemisphere, which gives them a "butterfly" appearance (Figure 3).

Figure 3

How do glioblastomas typically appear anatomically and on imaging?

Histology

Grossly, these tumors are usually large and solitary. As high-grade anaplastic tumors, histologically, they demonstrate marked nuclear atypia and mitotic activity. Furthermore, areas of pseudopalisading necrosis are seen, which means that cells are arranged in a "picket-fence" pattern surrounding a necrotic core (Figure 4).

QUIZ

Tap image for quiz

Figure 4

Presentation and Prognosis

Like all brain neoplasms, the presentation largely depends on the location of the tumor. Seizures are common; because of rapid growth, so are focal deficits such as paresis. Headaches do not necessarily follow the oft-described "brain tumor headache" pattern and may be clinically indistinguishable from tension headaches in a number of patients.

Unsurprisingly, glioblastomas are rapidly fatal; median survival from diagnosis is 15 months. The 5-year survival rate is <5%. Debulking is the gold-standard treatment, but because the tumor is so invasive, the unresected portion is treated with radiation and chemotherapy.

What Are Oligodendrogliomas?

Oligodendrogliomas are tumors arising from oligodendrocytes, which are responsible for myelin production in the CNS.

CLINICAL CORRELATION

Oligodendrocytes are the CNS equivalent of Schwann cells in the peripheral nervous system. Both are responsible for myelin production, but they achieve this in different ways. An oligodendrocyte has many tentacles known as processes that extend out to multiple nerve cells, forming a coat of myelin. This allows a single oligodendrocyte to myelinate up to 50 different axons! In contrast, Schwann cells myelinate one individual nerve by wrapping its cell bodies around the axon.

Incidence and Location

The annual incidence of oligodendrogliomas is 0.27 cases per 100,000 people, and they account for about 10% of primary brain tumor gliomas. These tumors typically occur in adults with a peak incidence at age 35-44 years. They can occur anywhere in the cerebral hemispheres, but they most commonly occur in the frontal and temporal lobes.

Histology

Histologically, oligodendrogliomas have a characteristic "fried-egg" appearance with a "chicken-wire" capillary pattern (Figure 5). The fried-egg description refers to cells with a round nuclei and clear cytoplasm—kind of like the yolk in the middle of the egg white. The chicken-wire capillary pattern refers to the numerous wisp-like blood vessels. These tumors also have areas of mucoid and cystic degeneration with hemorrhage and calcifications on histology. Molecularly, there is a strong association with 1p and 19q deletions, which are also associated with a more favorable prognosis.

QUIZ

Tap image for quiz

Figure 5

What is the typical histological finding in oligodendroglioma?

What is the typical histological finding in oligodendroglioma?

Presentation

Approximately, 66% of patients with oligodendrogliomas present with seizures. Headaches and focalizing deficits may also be present. These tumors are most commonly found in the frontal and temporal lobes. The frontal lobe is important for controlling motor movements and personality. Accordingly, tumors in the frontal lobe manifest as unilateral muscle weakness and personality changes. Temporal lobe masses do not usually present with noticeable symptoms other than seizures and language deficits.

Prognosis

The first-line treatment for surgically accessible oligodendrogliomas is resection. For inaccessible tumors, chemotherapy and radiation are viable treatment options. The 1-, 5-, and 10-year survival rates for oligodendrogliomas are 93.8%, 79.1%, and 62.6%, respectively. Anaplastic oligodendrogliomas (WHO grade III) have worse prognoses, with 1-, 5-, and 10-year survival rates of 80.6%, 50.7%, and 37.3%, respectively.

What Are Ependymomas?

As the name suggests, this tumor arises from the ependymal cells. Ependymal cells have an important role in cerebrospinal fluid (CSF) production and regulation. They are found throughout the brain and spinal cord.

Incidence and Location

Ependymomas account for only 2%-3% of primary brain tumors in adults. However, they are the third most common form of childhood brain and spine tumors, with 30% being diagnosed in children younger than 3 years. In children, these tumors typically arise from the floor of the fourth ventricle, causing an obstructive hydrocephalus. In adults, these tumors typically occur in the spinal cord. Spinal cord tumors are associated with a better prognosis because they are easier to remove than those occurring within the ventricles.

Histology

Histologically, we can see rod-shaped basal bodies and perivascular pseudorosettes, which are spiral clusters of cells around vascular structures that resemble roses. These are formed as cell processes of ependymal cells extend toward blood vessels and surround them, creating an area devoid of nuclei (Figure 6). Rod-shaped basal bodies are modified centrioles. These tumors also form ependymal rosettes, which are circumferentially arranged ependymal tumor cells around an empty lumen thought to reflect the process ependymal cells take to form the cerebral ventricles!

QUIZ

Tap image for quiz

Figure 6

CLINICAL CORRELATION

What are rosettes vs pseudorosettes? A rosette is a spoke-wheel arrangement of cells surrounding a central core, which is an empty-appearing lumen or space containing cytoplasmic processes. In a pseudorosette, cells also arrange in a spoke-wheel pattern but have tapered processes that extend out, surrounding a central hub that is not formed by the tumor. For example, the perivascular pseudorosettes of ependymomas have a central hub composed of a vessel.

Presentation

Because these tumors typically arise from the floor of the fourth ventricle in children, obstructive hydrocephalus can develop. Signs of obstructive hydrocephalus include headache, nausea, vomiting, gait difficulties, macrocephaly, and lethargy. Tumors arising in the spinal cord, which are more commonly seen in adults, tend to course with dysesthesias, spastic paresis below the level of the lesion, and gait deficits. Because CSF seeding or "drop metastases" could occur, any patient with ependymoma must have imaging of the whole neuraxis.

Where do ependymomas typically occur in children?

Prognosis

The greatest prognostic predictor for patients with an ependymoma is the extent of resection. For patients with complete resection, the overall 5-year progression-free survival rate is about 70%, compared with 30%-40% for those with only partially resected tumors. Targeted radiation after surgery is currently the standard of care.

TRUCTOR NOTE

Many ependymomas are in critical locations that are not amenable to surgical resection. Therefore, though they may be considered WHO grade 2, the prognosis is worse because of the lack of accessibility for removal.

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CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to TC, what are the most likely causes of his seizure and headaches?

A new-onset seizure in an older patient usually indicates a serious underlying cause, such as a stroke, intracranial hemorrhage, or tumor. TC's scan was performed quickly, and you summarize the result for TC and BC. "I need to tell you some bad news. The scan shows a large mass that looks like a tumor. We'll need to do additional tests to know what kind of tumor and how to treat it, but we know already it is a serious condition. The positive news is that TC has woken up and is doing better now. We'll arrange for TC to be seen by the neurosurgical specialists to define the treatment plan." TC and BC are shaken by this bad news, and you step out to give them some privacy to process what they've learned.

Summary

- Gliomas are tumors arising from glial cells, including astrocytes, oligodendrocytes, and ependymal cells.
- Pilocytic astrocytomas, the most common glial tumor in children, are typically benign; they exhibit Rosenthal fibers on histology.
- Infiltrating astrocytomas are WHO grade II tumors that more commonly occur in adults and invade surrounding tissue; on histology, they show hypercellularity with possible areas of microcytosis and calcifications.
- Oligodendrogliomas are tumors of oligodendrocytes that ordinarily myelinate axons in the CNS; oligodendrogliomas have the appearance of fried-egg cells and chicken-wire capillaries on histology.
- Ependymomas tend to occur in the floor of the fourth ventricle in children, causing an obstructive hydrocephalus; in adults, ependymomas usually occur in the spinal cord, creating compressive symptoms.
- Ependymomas have perivasculature pseudorosettes, ependymal rosettes, and basal bodies on histology.

Review Questions

1. **A 7-year-old previously healthy boy is brought to the emergency department by paramedics after his friend saw him having a seizure. Head computed tomography (CT) showed a mass in the left temporal**